2、新建IAR_S32K344的Autosar最小系统工程

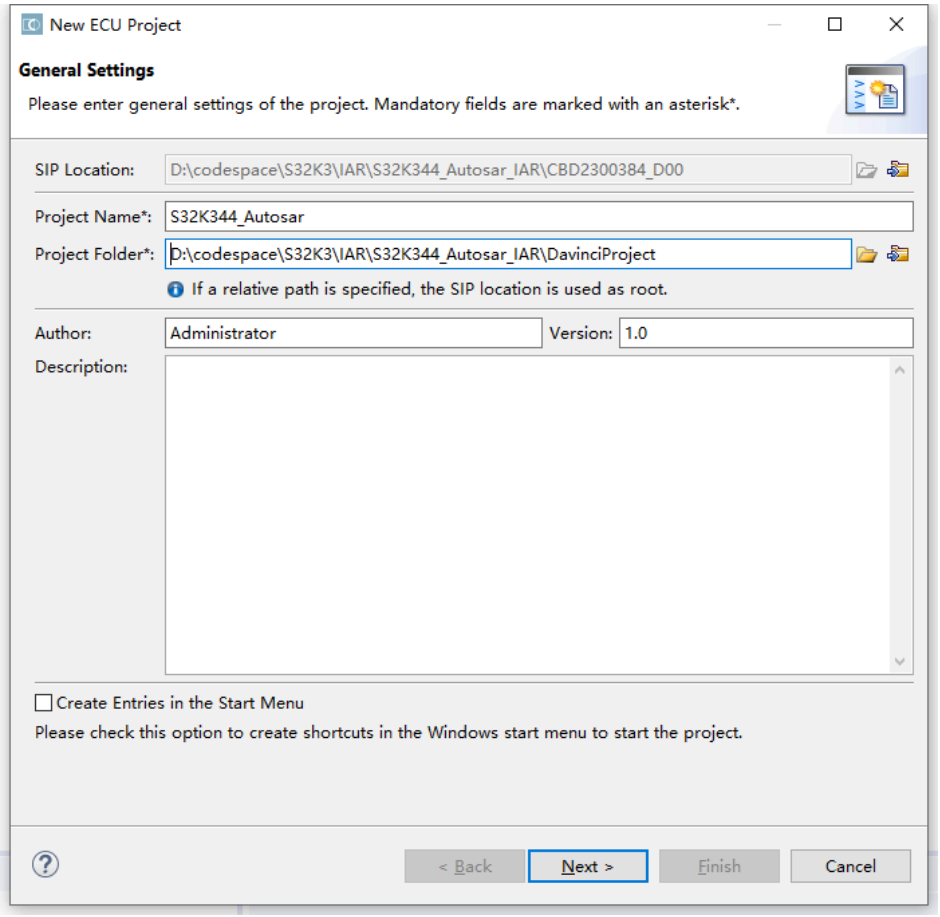
卖家'漩涡遨游aa' 专属资料，非在漩涡遨游aa 购买资料，不提供答疑(一次购买仅提供有限次数10次答疑)

这里SIP路径为D:\codespace\S32K3\IAR\S32K344_Autosar_IAR，后续此路径为工程的基础路径

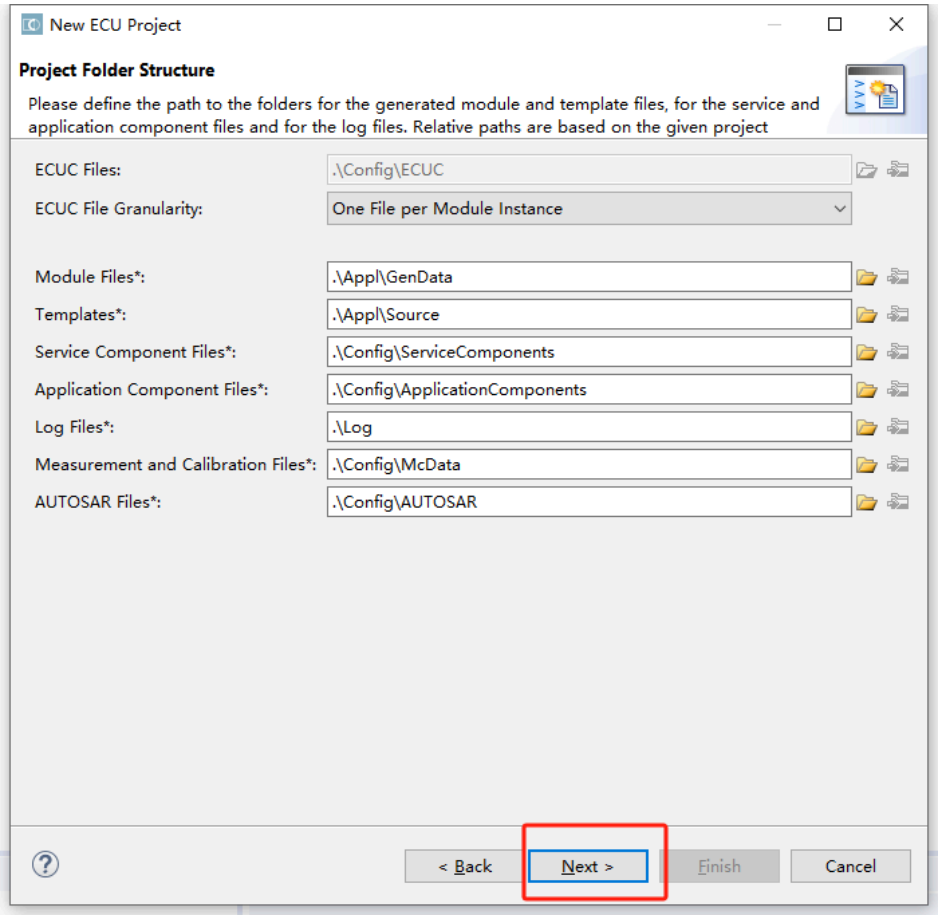
一、在SIP包中打开DaVinciCFG.exe新建davinci工程

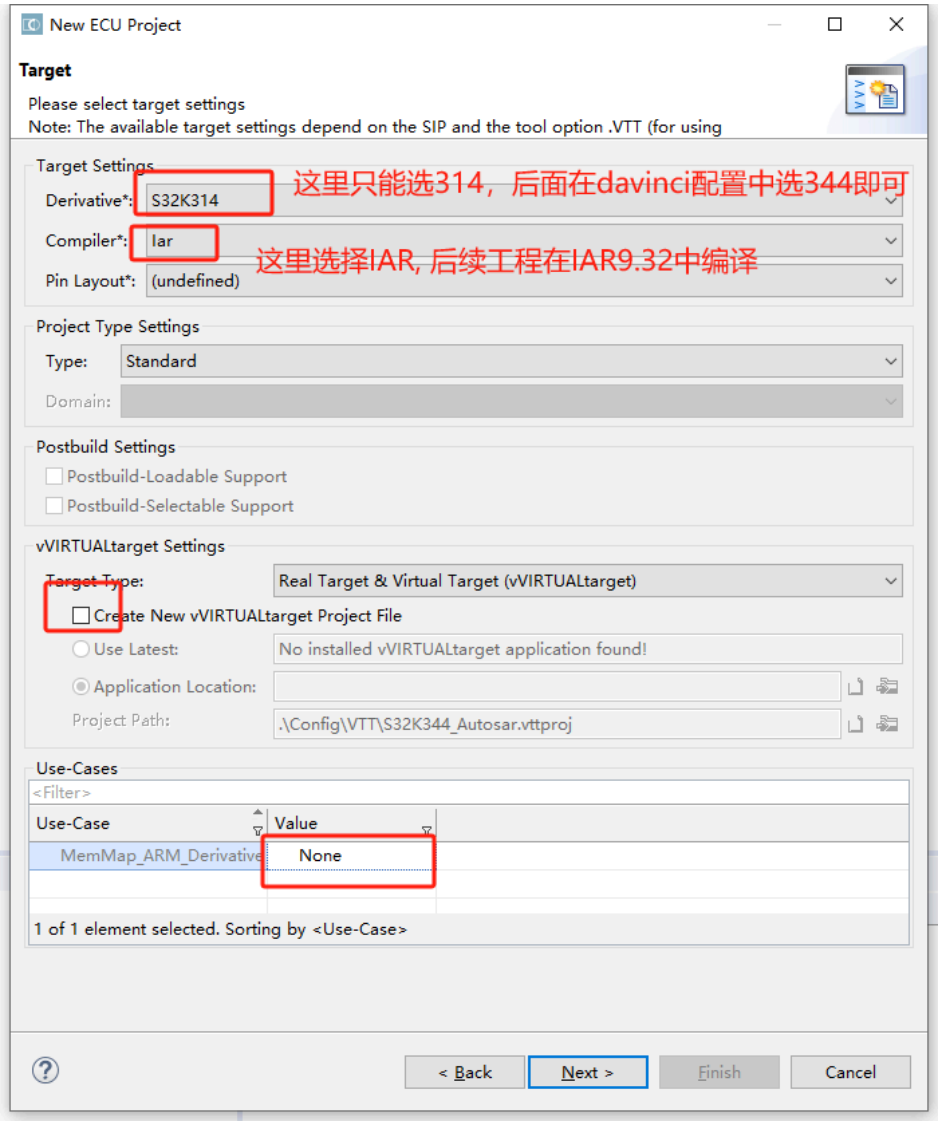
SIP包为网上咸鱼获取的，CBD2300384_D00，打开CBD2300384_D00\DaVinciConfigurator\Core\DaVinciCFG.exe

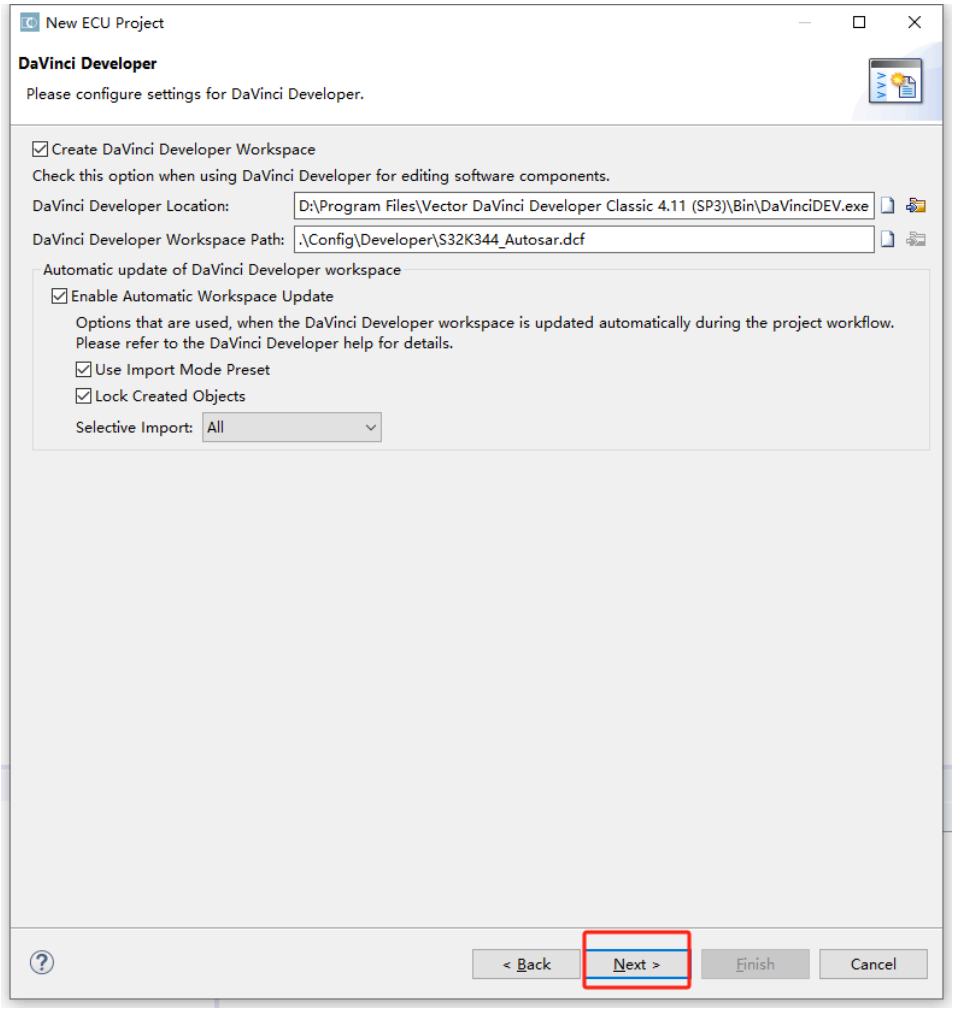
点击Flie→New ECU Project.... ， 填写好工程名S32K344_Autosar和工程路径

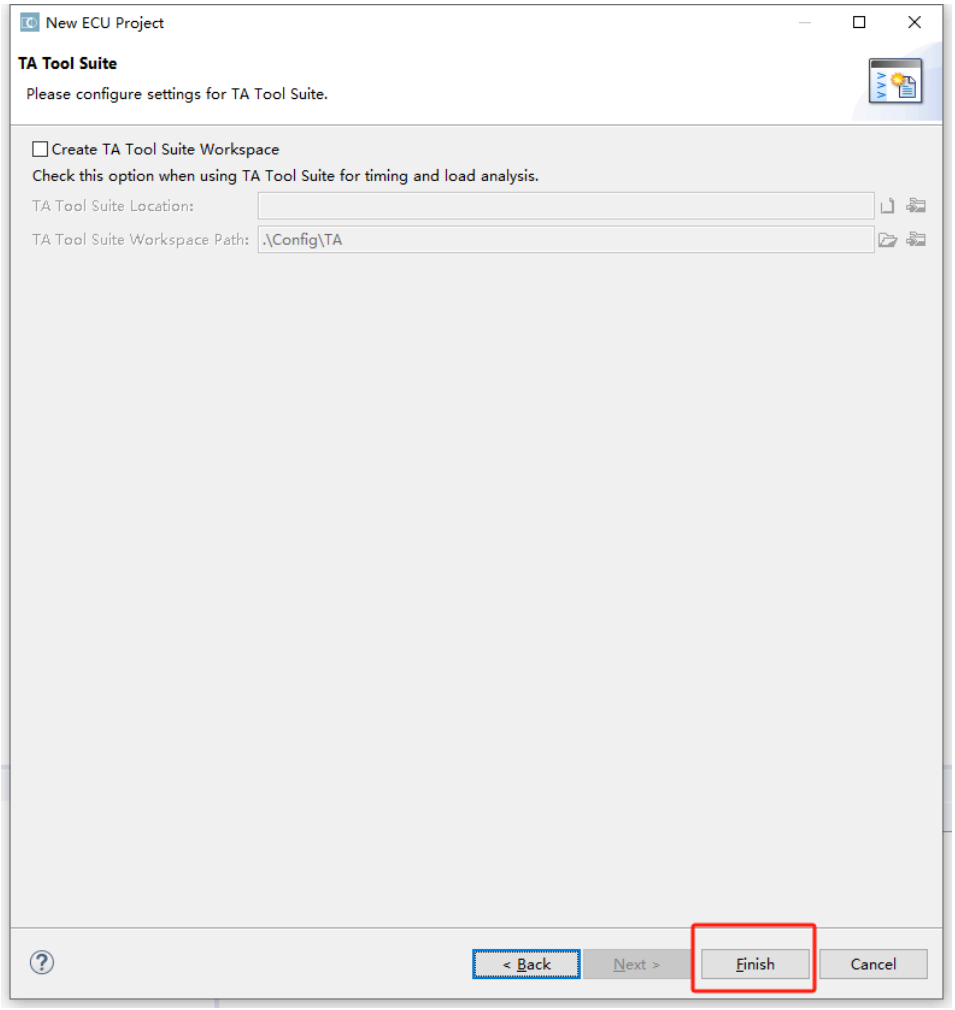


next


New ECU Project
Project Folder Structure
Please define the path to the folders for the generated module and template files, for the service and application component files and for the log files. Relative paths are based on the given project
ECUC Files:
.\Config\ECUC
ECUC File Granularity:
One File per Module Instance
Module Files*:
.\Appl\GenData
Templates*:
.\Appl\Source
Service Component Files*:
.\Config\ServiceComponents
Application Component Files*:
.\Config\ApplicationComponents
Log Files*:
.\Log
Measurement and Calibration Files*:
.\Config\McData
AUTOSAR Files*:
.\Config\AUTOSAR
< Back
Next >
Finish
Cancel


New ECU Project
Target
Please select target settings
Note: The available target settings depend on the SIP and the tool option .VTT (for using
Target Settings
Derivative*: S32K314
这里只能选314，后面在davinci配置中选344即可
Compiler*: Iar
这里选择IAR, 后续工程在IAR9.32中编译
Pin Layout*: (undefined)
Project Type Settings
Type: Standard
Domain:
Postbuild Settings
Postbuild-Loadable Support
Postbuild-Selectable Support
vVIRTUALtarget Settings
Target Type: Real Target & Virtual Target (vVIRTUALtarget)
Create New vVIRTUALtarget Project File
Use Latest: No installed vVIRTUALtarget application found!
Application Location:
Project Path: .\Config\VTT\S32K344_Autosar.vttproj
Use-Cases
<Filter>
Use-Case
Value
MemMap_ARM_Derivative
None
1 of 1 element selected. Sorting by <Use-Case>
< Back
Next >
Finish
Cancel

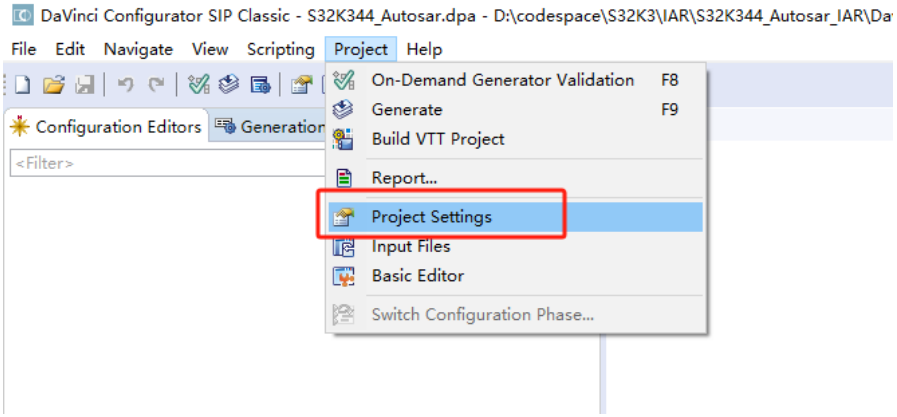

New ECU Project
DaVinci Developer
Please configure settings for DaVinci Developer.
Create DaVinci Developer Workspace
Check this option when using DaVinci Developer for editing software components.
DaVinci Developer Location:
D:\Program Files\Vector DaVinci Developer Classic 4.11 (SP3)\Bin\DaVinciDEV.exe
DaVinci Developer Workspace Path:
.\Config\Developer\S32K344_Autosar.dcf
Automatic update of DaVinci Developer workspace
Enable Automatic Workspace Update
Options that are used, when the DaVinci Developer workspace is updated automatically during the project workflow.
Please refer to the DaVinci Developer help for details.
Use Import Mode Preset
Lock Created Objects
Selective Import:
All
< Back
Next >
Finish
Cancel

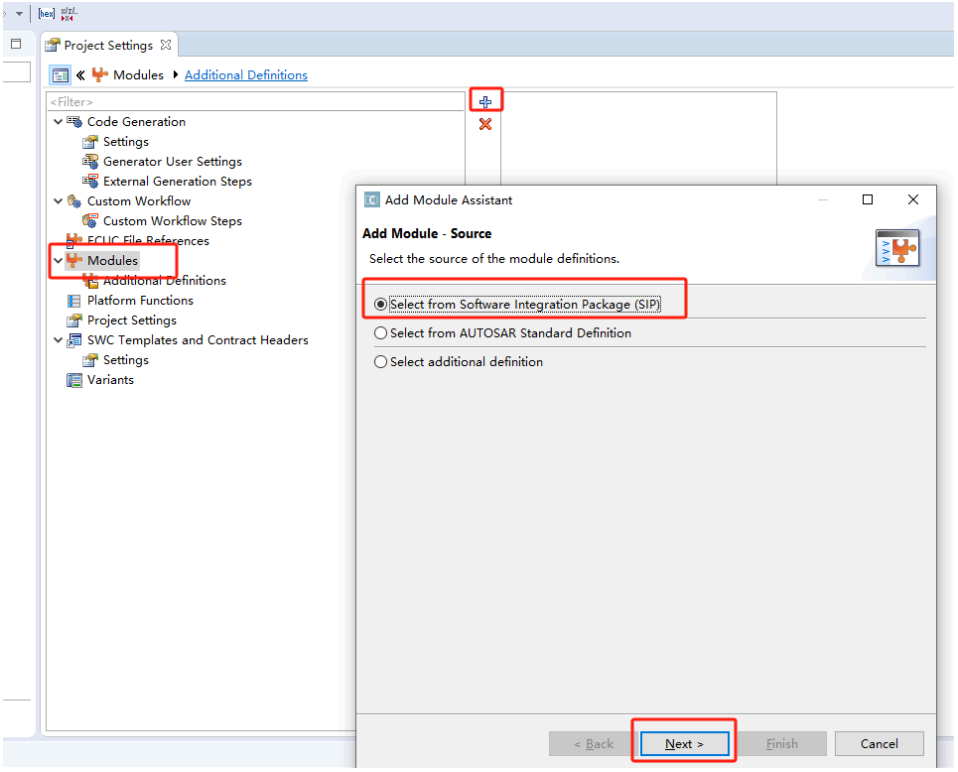


二、添加最小系统需要的模块

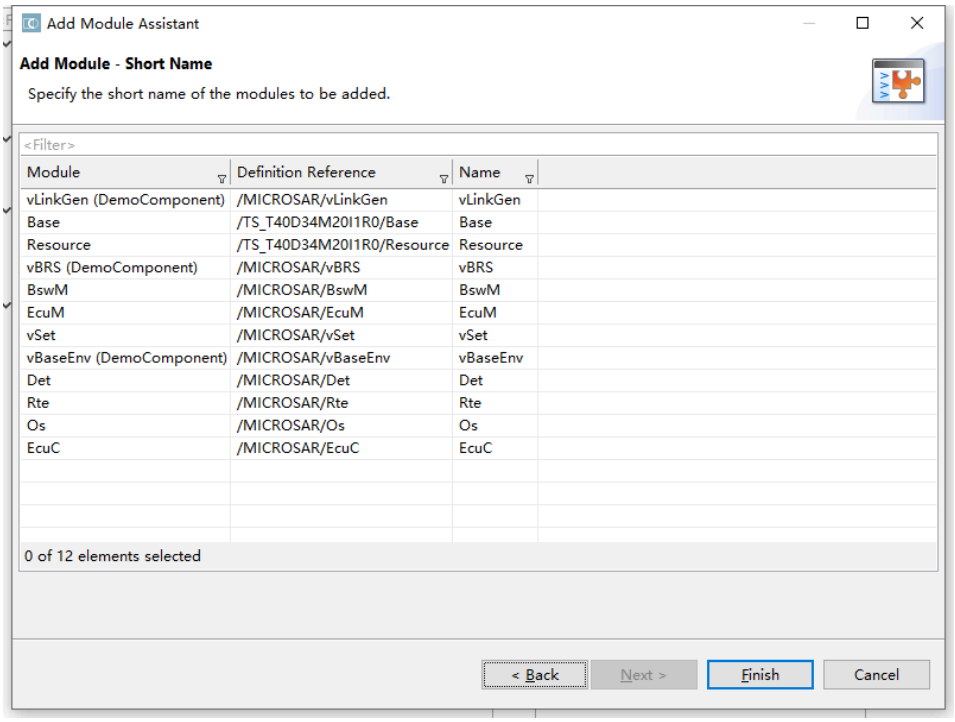
打开 S32K344_Autosar_IAR\DavinciProject\S32K344_Autosar.dpa

2.1、添加基础模块



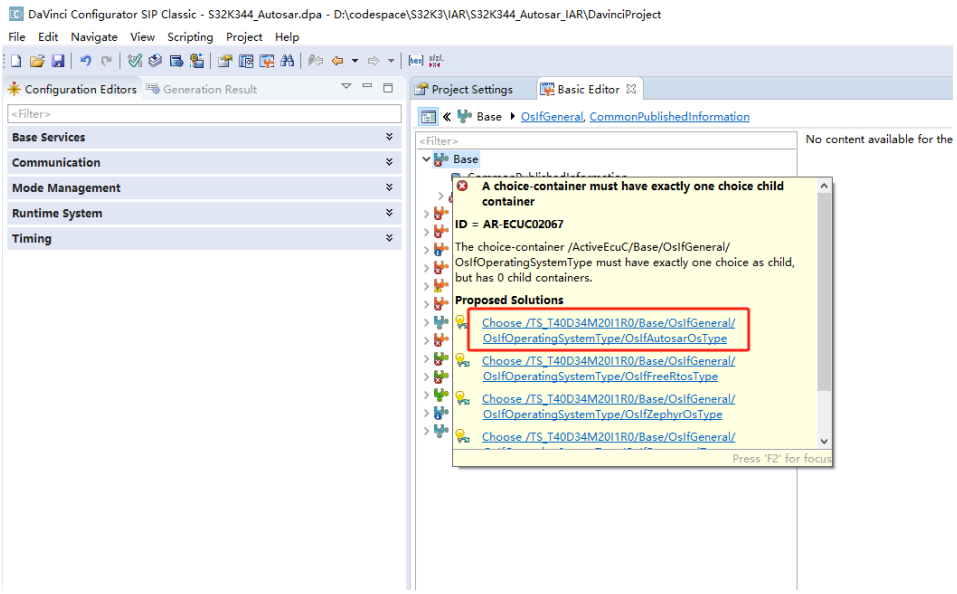


添加Base、Mcu、Resource(这三个是MCAL模块)、BswM、Det、EcuC、EcuM、Os、Rte、vBRS、vBaseEnv、vLinkGen、vSet

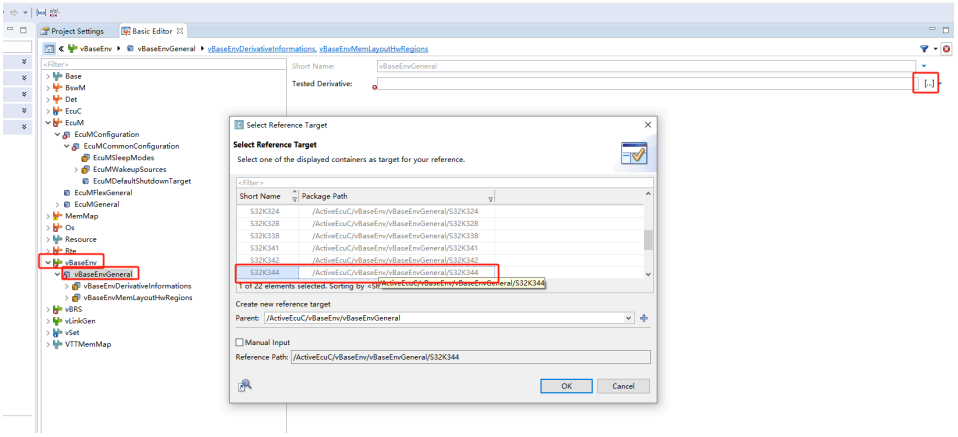


2.2、Base

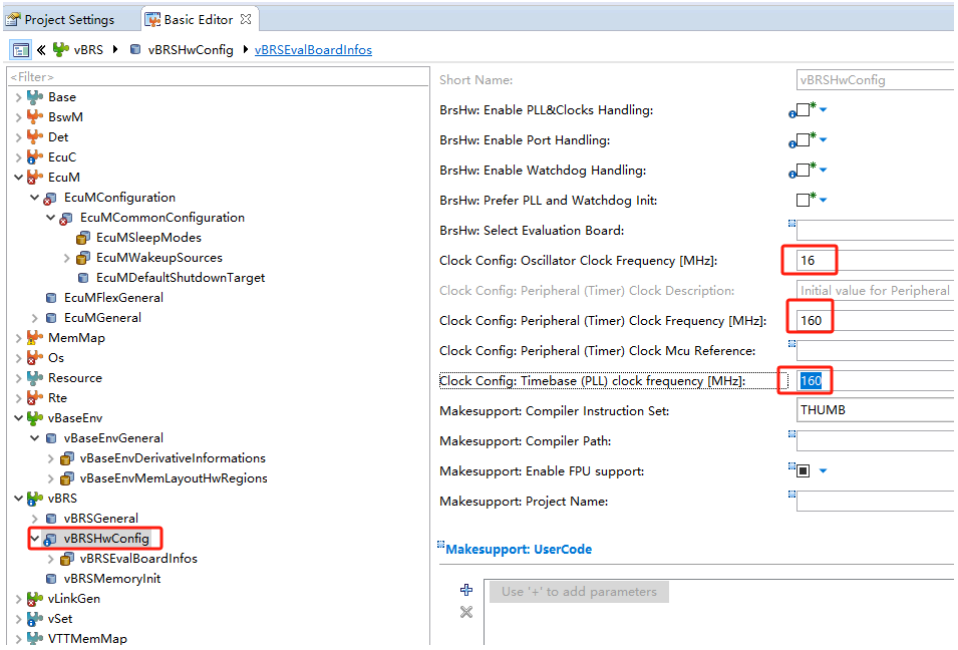
选择autosar os的



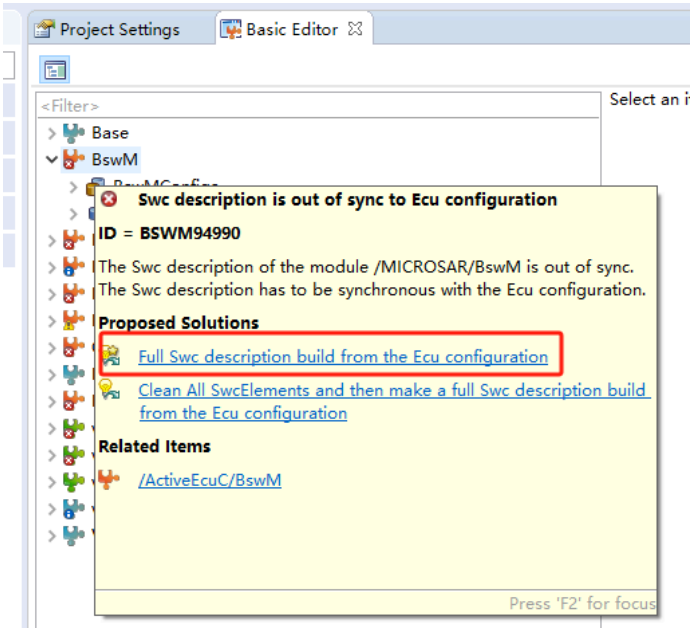
2.3、vBaseEnv



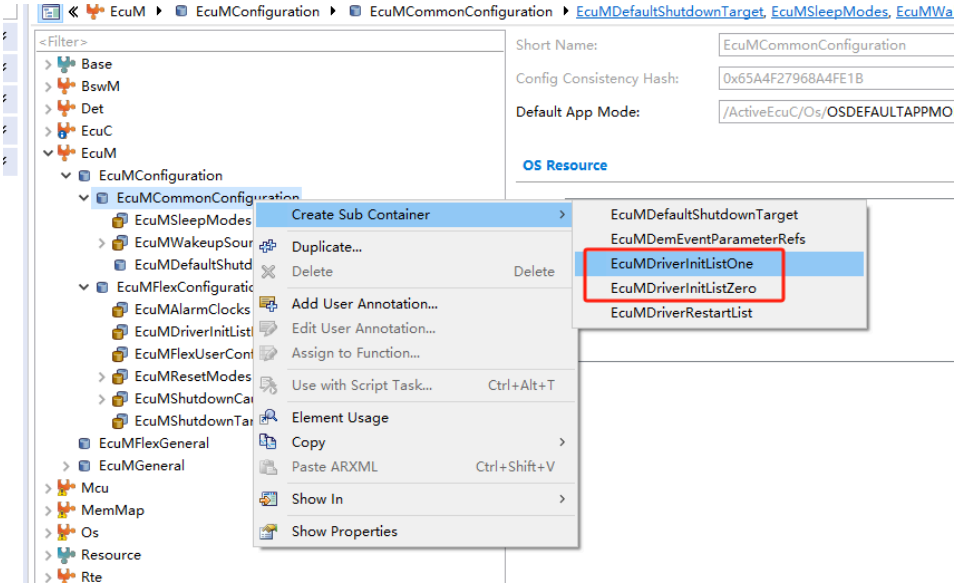
2.4、vBRS



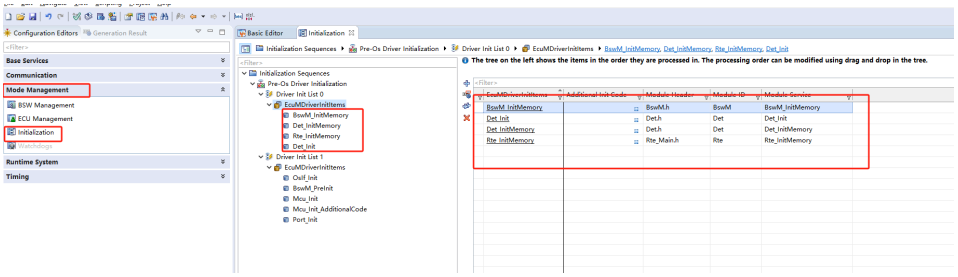
2.5、BswM



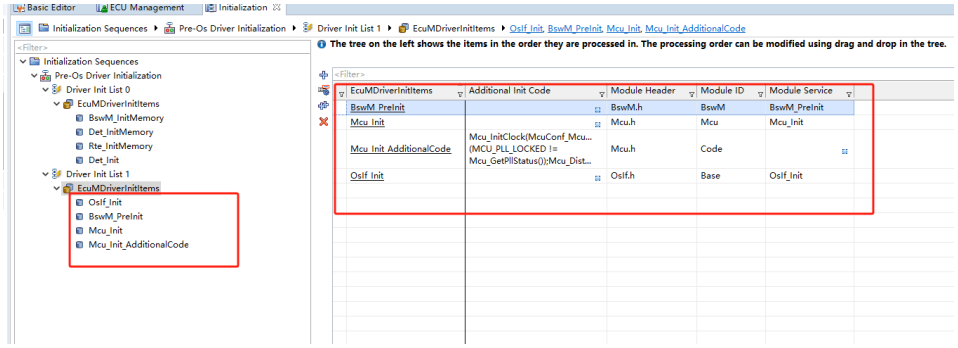
添加EcuMDriverInitListOne 和 EcuMDriverInitListZero



EcuMDriverInitListZero 按照顺序添加以下Init！

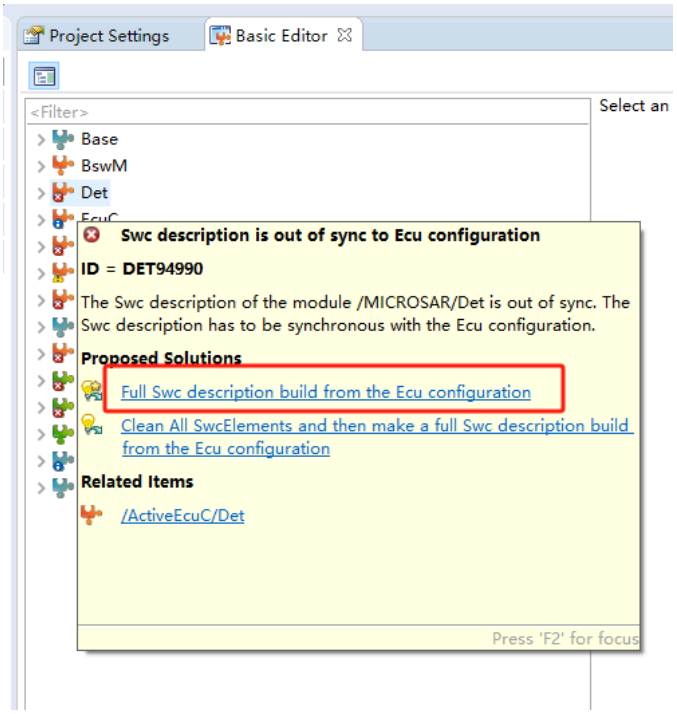


EcuMDriverInitListOne 按照顺序添加以下Init！！

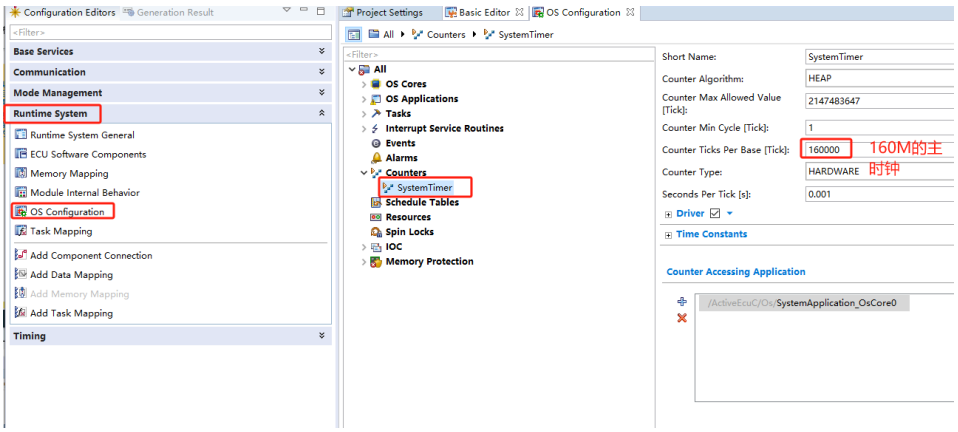


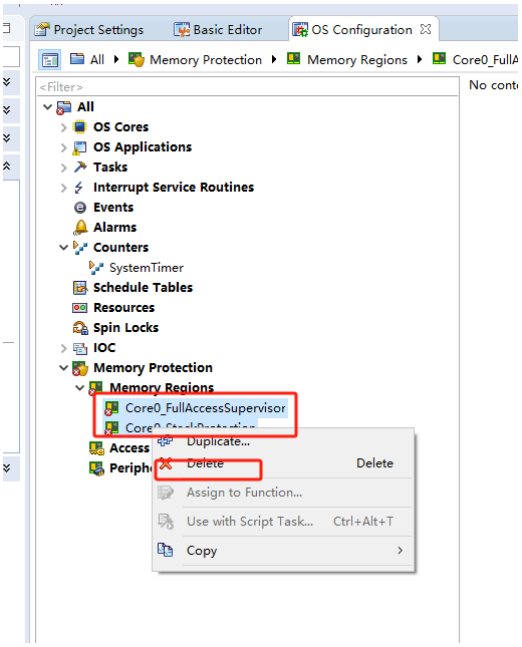
其中mcu 额外的代码如下
Mcu_InitClock(McuConf_McuClockSettingConfig_McuClockSettingConfig_0);while (MCU_PLL_LOCKED != Mcu_GetPllStatus());Mcu_DistributePllClock();Mcu_SetMode(McuConf_McuModeSettingConf_McuModeSettingConf_0);

2.6、Det

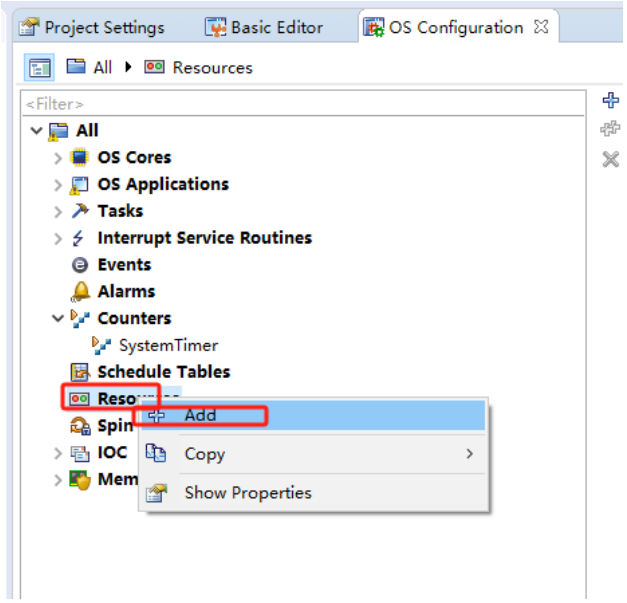

Project Settings
Basic Editor
<Filter>
Select an
Base
BswM
Det
Swc description is out of sync to Ecu configuration
ID = DET94990
The Swc description of the module /MICROSAR/Det is out of sync. The Swc description has to be synchronous with the Ecu configuration.
Proposed Solutions
Full Swc description build from the Ecu configuration
Clean All SwcElements and then make a full Swc description build from the Ecu configuration
Related Items
/ActiveEcuC/Det
Press 'F2' for focus

2.7、Os

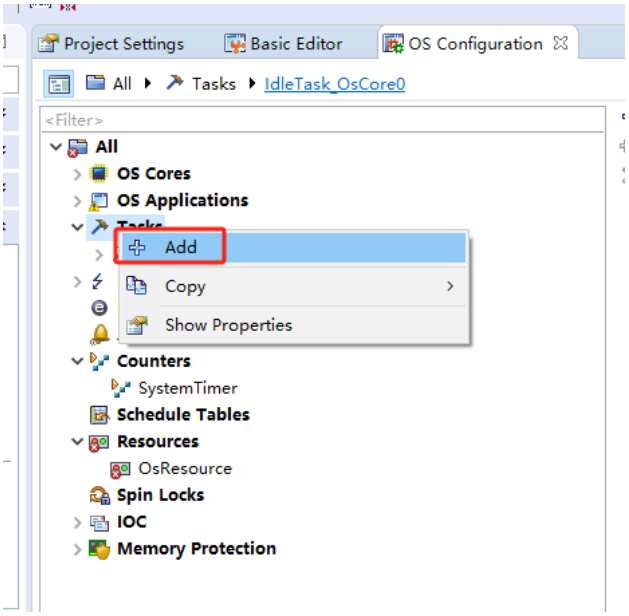

Configuration Editors
Generation Result
Base Services
Communication
Mode Management
Runtime System
Runtime System General
ECU Software Components
Memory Mapping
Module Internal Behavior
OS Configuration
Task Mapping
Add Component Connection
Add Data Mapping
Add Memory Mapping
Add Task Mapping
Timing
Project Settings
Basic Editor
OS Configuration
All
Counters
SystemTimer
All
OS Cores
OS Applications
Tasks
Interrupt Service Routines
Events
Alarms
Counters
SystemTimer
Schedule Tables
Resources
Spin Locks
IOC
Memory Protection
Short Name:
SystemTimer
Counter Algorithm:
HEAP
Counter Max Allowed Value [Tick]:
2147483647
Counter Min Cycle [Tick]:
1
Counter Ticks Per Base [Tick]:
160000
160M的主时钟
Counter Type:
HARDWARE
Seconds Per Tick [s]:
0.001
Driver
Time Constants
Counter Accessing Application
/ActiveEcuC/Os/SystemApplication_OsCore0



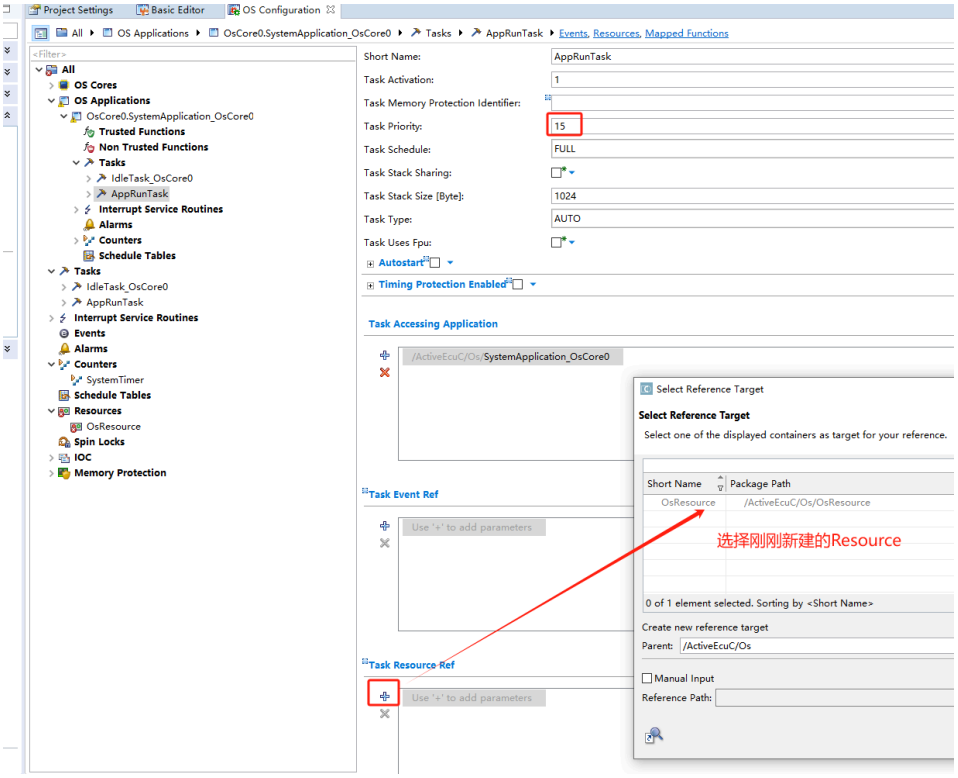
新建一个Resource

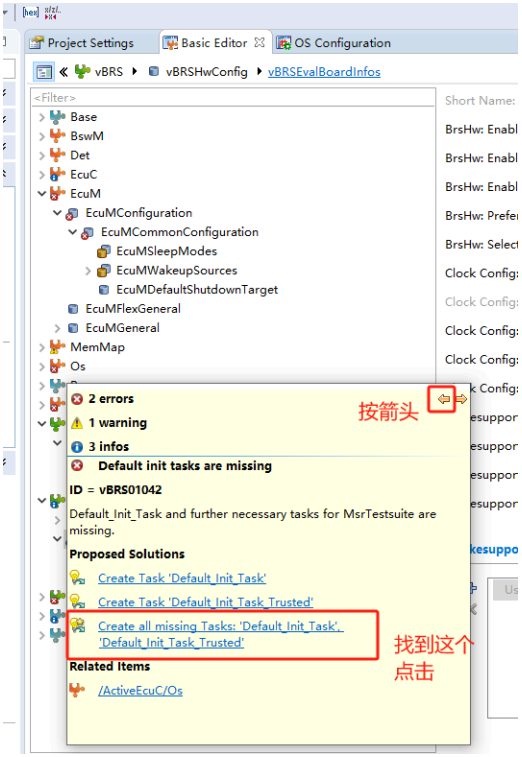


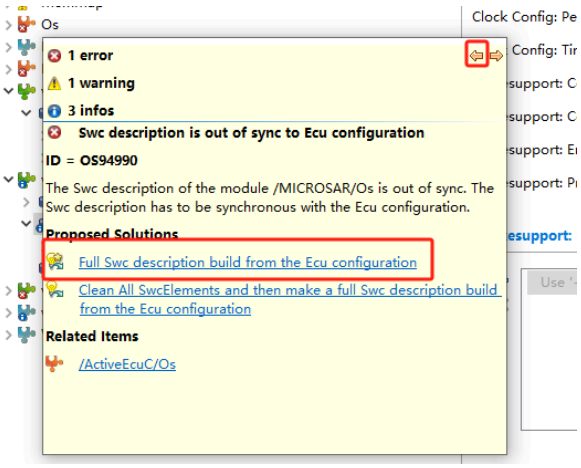
新建一个task,



改名为AppRunTask， 用于后续放BswM等一些的runnable

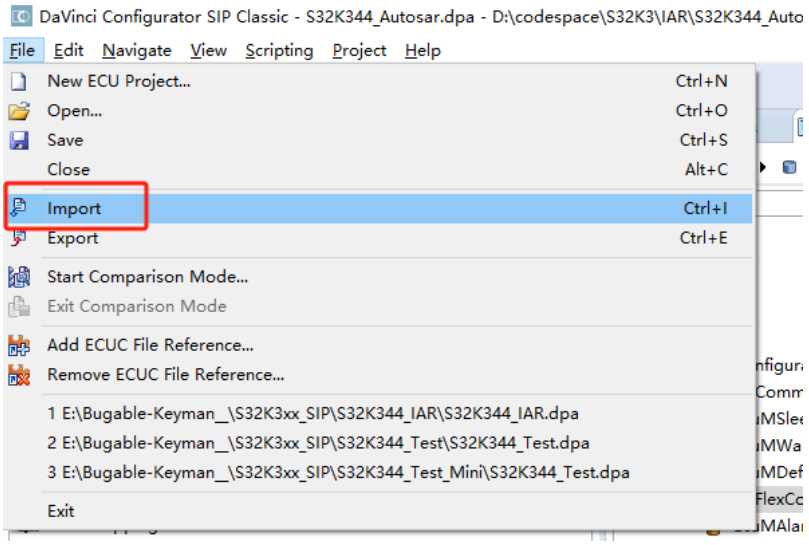




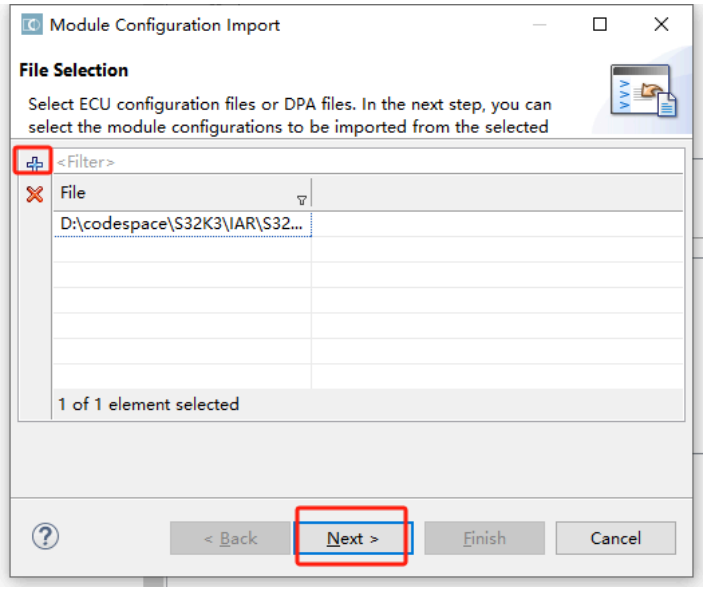


2.8、导入EB生成的Mcu模块

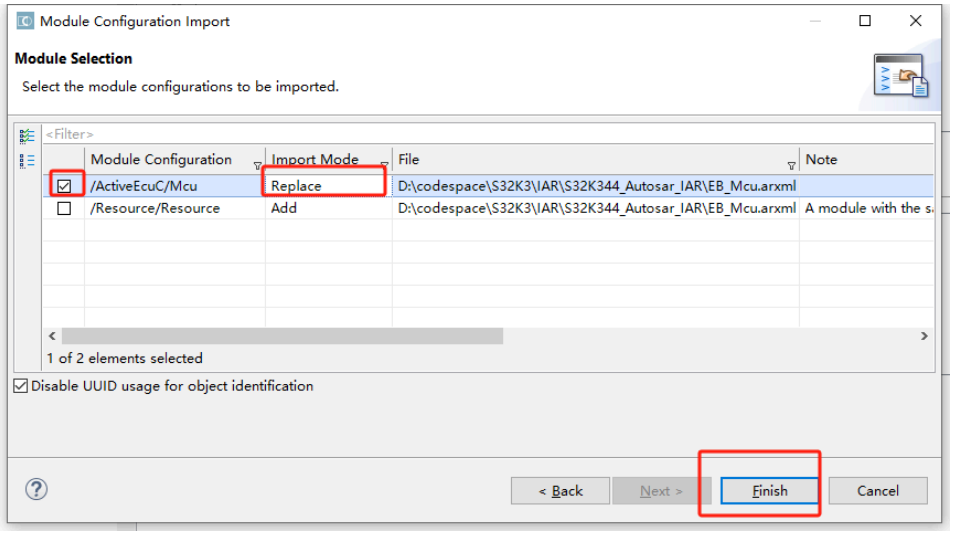
由于Mcu配置较为复杂，且配错了在davinci也不方便检查，可以看文档EB配置S32K344的mcal 的三、Mcu配置章节，里面会导出arxml用于这里的导入



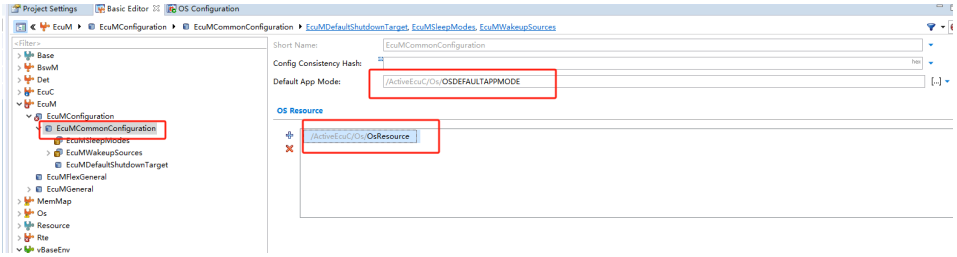
添加附件包中的EB_Mcu.arxml

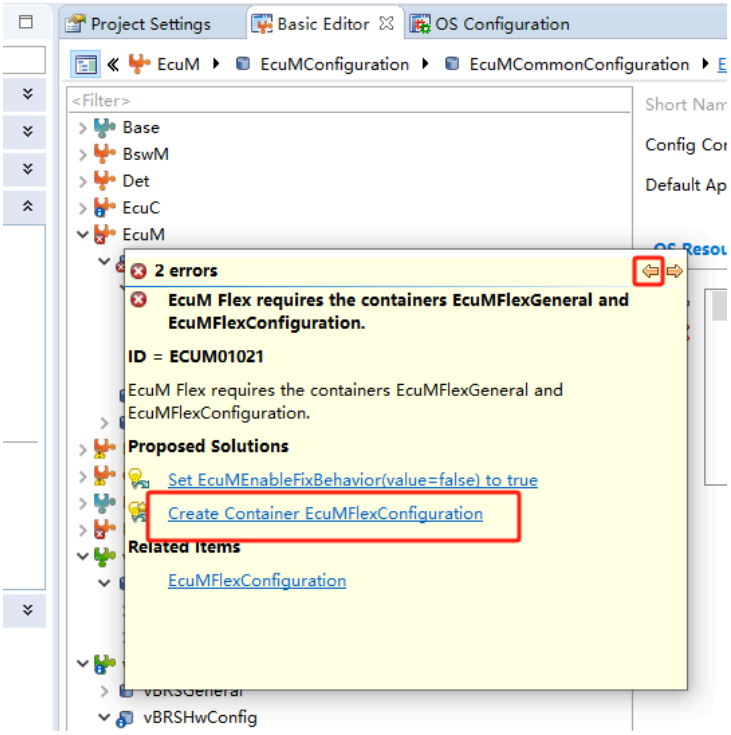


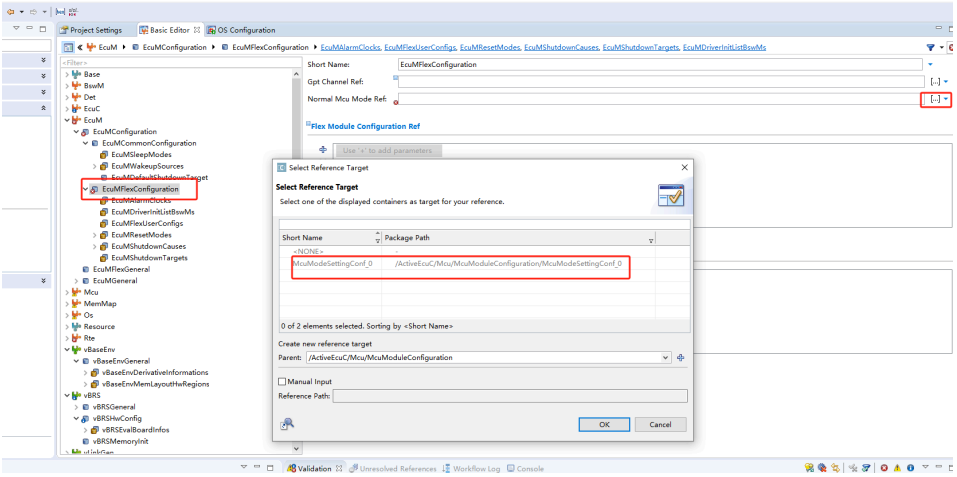
只选择Mcu, 且选replace

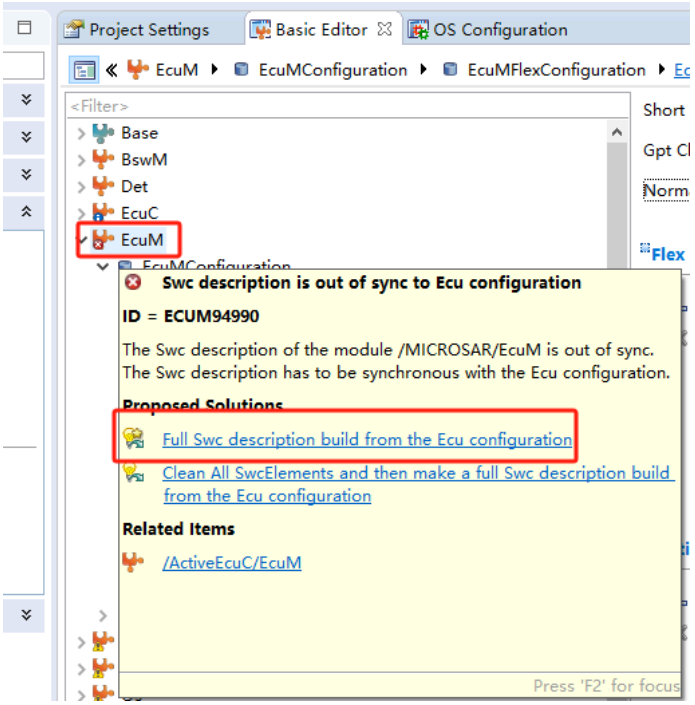


2.9、EcuM

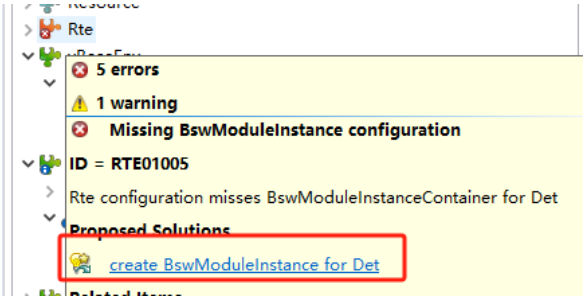

Project Settings
Basic Editor
OS Configuration
EcuM
EcuMConfiguration
EcuMCommonConfiguration
Short Name:
Config Consistency Hash:
Default App Mode:
/ActiveEcuC/Os/OSDEFAULTAPPMODE
OS Resource
/ActiveEcuC/Os/OsResource

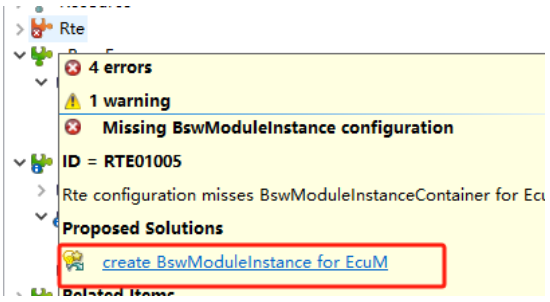

Project Settings
Basic Editor
OS Configuration
EcuM
EcuMConfiguration
EcuMCommonConfiguration
<Filter>
Base
BswM
Det
EcuC
EcuM
2 errors
EcuM Flex requires the containers EcuMFlexGeneral and EcuMFlexConfiguration.
ID = ECUM01021
EcuM Flex requires the containers EcuMFlexGeneral and EcuMFlexConfiguration.
Proposed Solutions
Set EcuMEnableFixBehavior(value=false) to true
Create Container EcuMFlexConfiguration
Related Items
EcuMFlexConfiguration
vBRSHwConfig

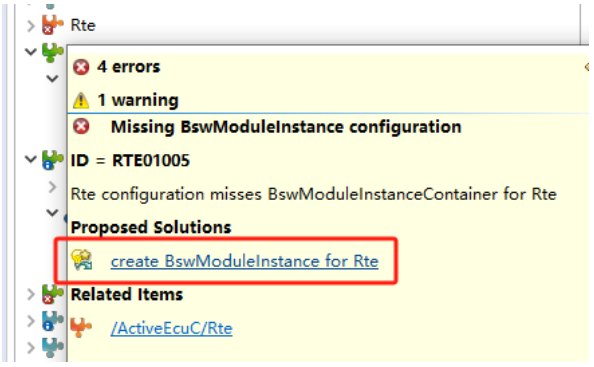

Project Settings
Basic Editor
OS Configuration
EcuM
EcuMConfiguration
EcuMFlexConfiguration
Short Name:
Gpt Channel Ref:
Normal Mcu Mode Ref:
Flex Module Configuration Ref
Select Reference Target
Select one of the displayed containers as target for your reference.
Short Name
Package Path
<NONE>
McuModeSettingConf_0
/ActiveEcuC/Mcu/McuModuleConfiguration/McuModeSettingConf_0
Create new reference target
Parent: /ActiveEcuC/Mcu/McuModuleConfiguration
Manual Input
Reference Path:
OK
Cancel

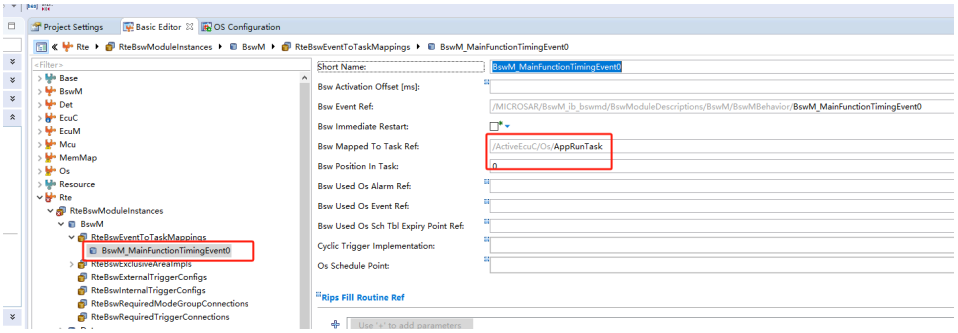

Project Settings
Basic Editor
OS Configuration
EcuM
EcuMConfiguration
EcuMFlexConfiguration
<Filter>
Base
BswM
Det
EcuC
EcuM
Swc description is out of sync to Ecu configuration
ID = ECUM94990
The Swc description of the module /MICROSAR/EcuM is out of sync.
The Swc description has to be synchronous with the Ecu configuration.
Proposed Solutions
Full Swc description build from the Ecu configuration
Clean All SwcElements and then make a full Swc description build from the Ecu configuration
Related Items
/ActiveEcuC/EcuM
Press 'F2' for focus

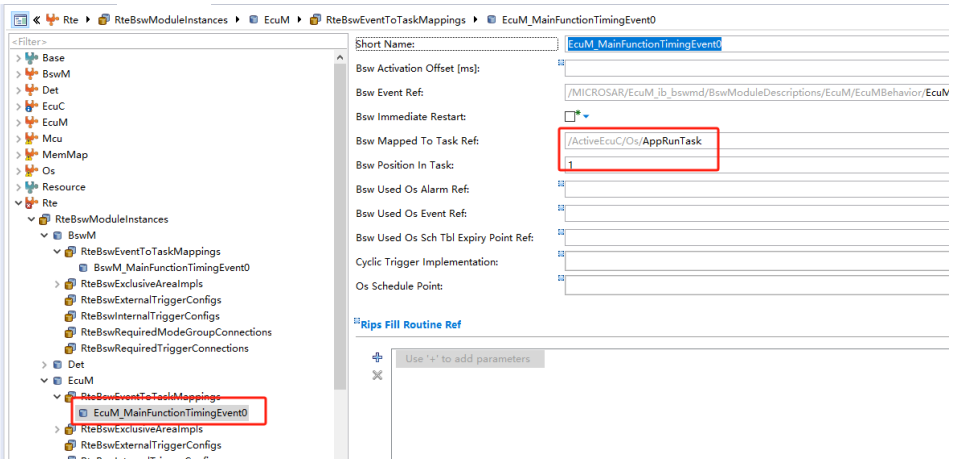
2.10、Rte

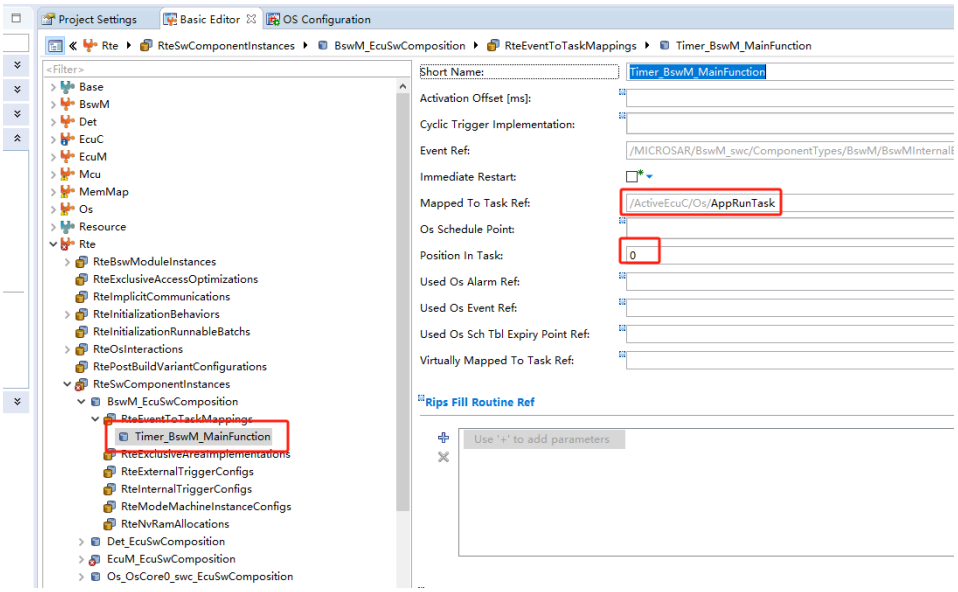

Rte
5 errors
1 warning
Missing BswModuleInstance configuration
ID = RTE01005
Rte configuration misses BswModuleInstanceContainer for Det
Proposed Solutions
create BswModuleInstance for Det

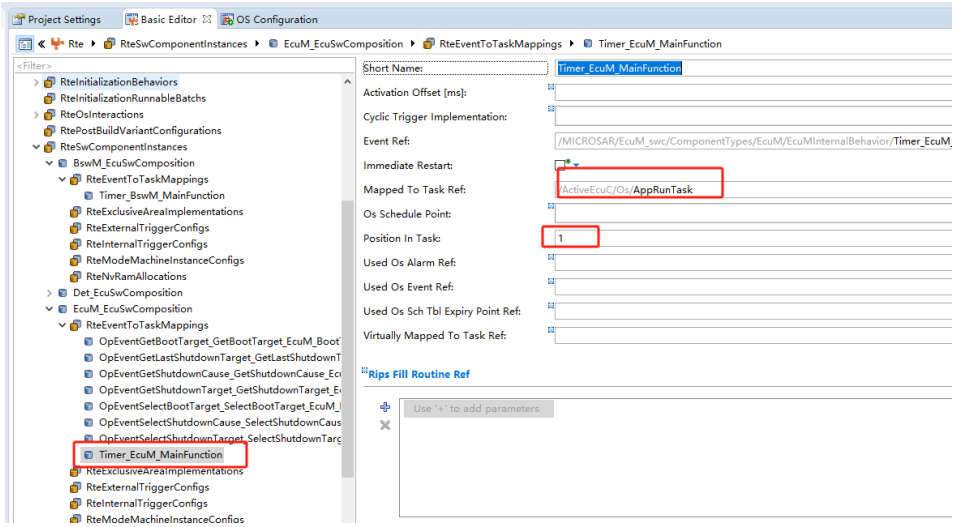

Rte
4 errors
1 warning
Missing BswModuleInstance configuration
ID = RTE01005
Rte configuration misses BswModuleInstanceContainer for Ecu
Proposed Solutions
create BswModuleInstance for EcuM


Rte
4 errors
1 warning
Missing BswModuleInstance configuration
ID = RTE01005
Rte configuration misses BswModuleInstanceContainer for Rte
Proposed Solutions
create BswModuleInstance for Rte
Related Items
/ActiveEcuC/Rte


Project Settings
Basic Editor
OS Configuration
Short Name:
BswM_MainFunctionTimingEvent0
Bsw Activation Offset [ms]:
Bsw Event Ref:
Bsw Immediate Restart:
Bsw Mapped To Task Ref:
/ActiveEcuC/Os/AppRunTask
Bsw Position In Task:
0
Bsw Used Os Alarm Ref:
Bsw Used Os Event Ref:
Bsw Used Os Sch Tbl Expiry Point Ref:
Cyclic Trigger Implementation:
Os Schedule Point:
Rips Fill Routine Ref

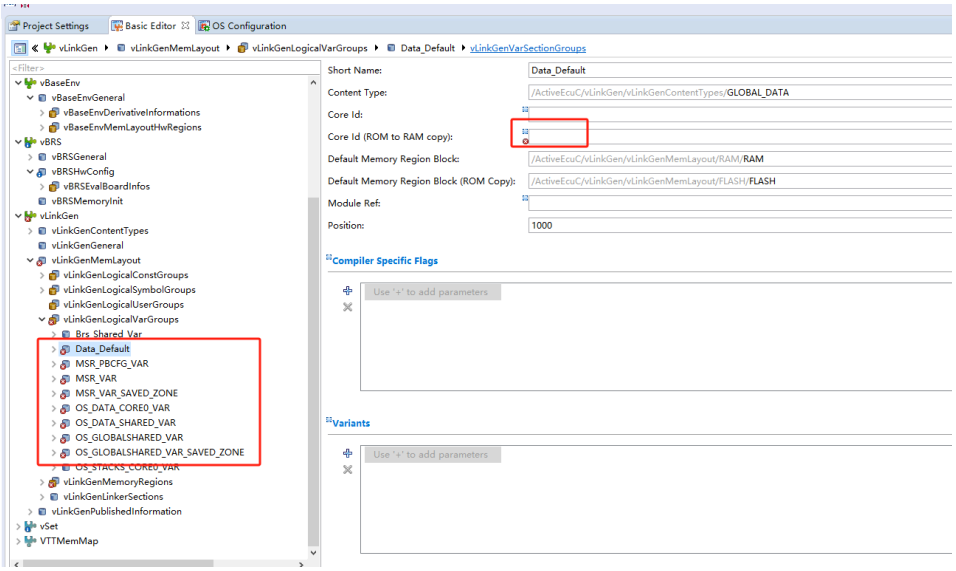

Short Name:
EcuM_MainFunctionTimingEvent0
Bsw Activation Offset [ms]:
Bsw Event Ref:
Bsw Immediate Restart:
Bsw Mapped To Task Ref:
/ActiveEcuC/Os/AppRunTask
Bsw Position In Task:
1
Bsw Used Os Alarm Ref:
Bsw Used Os Event Ref:
Bsw Used Os Sch Tbl Expiry Point Ref:
Cyclic Trigger Implementation:
Os Schedule Point:
Rips Fill Routine Ref
Use '+' to add parameters


Project Settings
Basic Editor
OS Configuration
Short Name:
Timer_BswM_MainFunction
Activation Offset [ms]:
Cyclic Trigger Implementation:
Event Ref:
Immediate Restart:
Mapped To Task Ref:
/ActiveEcuC/Os/AppRunTask
Os Schedule Point:
Position In Task:
0
Used Os Alarm Ref:
Used Os Event Ref:
Used Os Sch Tbl Expiry Point Ref:
Virtually Mapped To Task Ref:
Rips Fill Routine Ref
Use '+' to add parameters

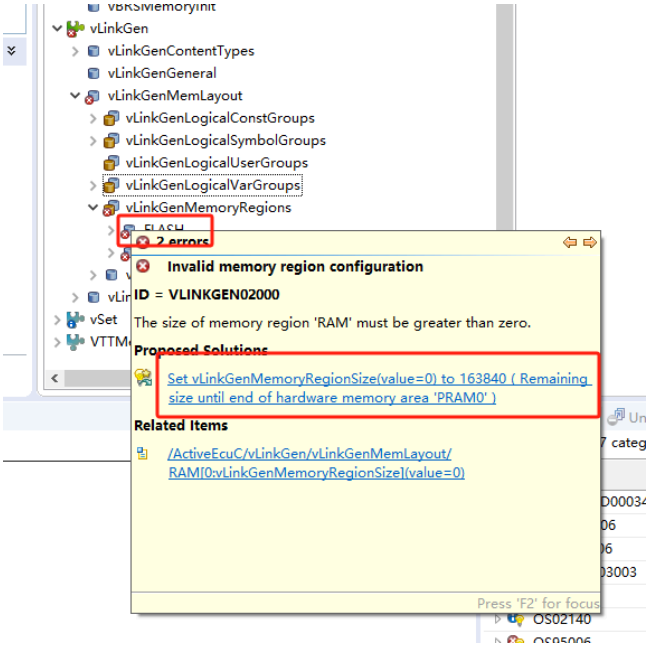


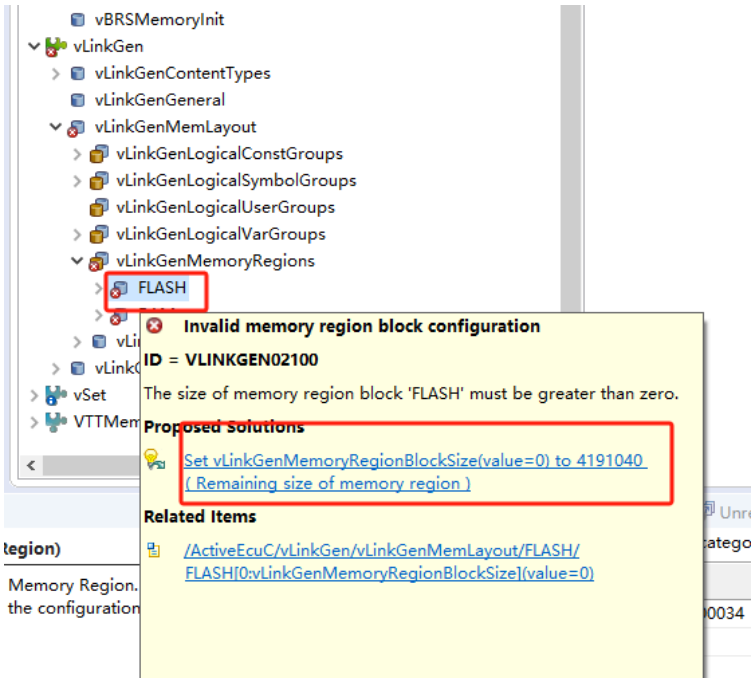
2.11、vLinkGen

这些的Core Id（ROM to RAM copy） 都填写0即可

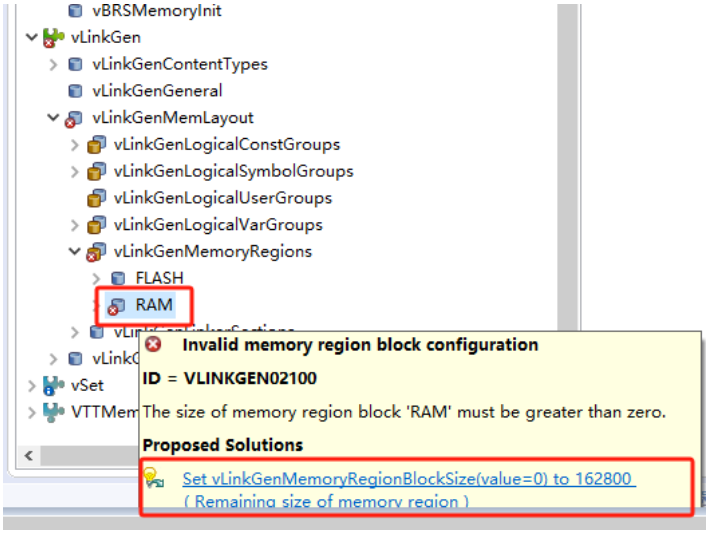


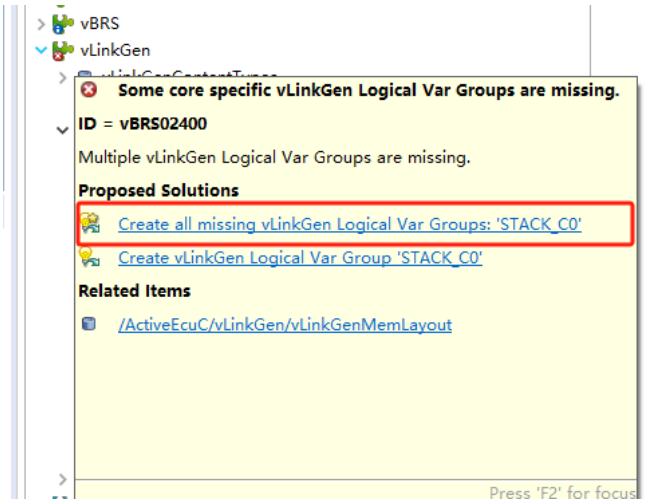
Flash 4M


vLinkGen
vLinkGenContentTypes
vLinkGenGeneral
vLinkGenMemLayout
vLinkGenLogicalConstGroups
vLinkGenLogicalSymbolGroups
vLinkGenLogicalUserGroups
vLinkGenLogicalVarGroups
vLinkGenMemoryRegions
2 errors
Invalid memory region configuration
ID = VLINKGEN02000
The size of memory region 'RAM' must be greater than zero.
Proposed Solutions
Set vLinkGenMemoryRegionSize(value=0) to 163840 (Remaining size until end of hardware memory area 'PRAM0')
Related Items
/ActiveEcuC/vLinkGen/vLinkGenMemLayout/RAM[0:vLinkGenMemoryRegionSize](value=0)
Press 'F2' for focus


vBRSMemoryInit
vLinkGen
vLinkGenContentTypes
vLinkGenGeneral
vLinkGenMemLayout
vLinkGenLogicalConstGroups
vLinkGenLogicalSymbolGroups
vLinkGenLogicalUserGroups
vLinkGenLogicalVarGroups
vLinkGenMemoryRegions
FLASH
vSet
VTTMem
Invalid memory region block configuration
ID = VLINKGEN02100
The size of memory region block 'FLASH' must be greater than zero.
Proposed Solutions
Set vLinkGenMemoryRegionBlockSize(value=0) to 4191040 (Remaining size of memory region)
Related Items
/ActiveEcuC/vLinkGen/vLinkGenMemLayout/FLASH/FLASH[0:vLinkGenMemoryRegionBlockSize](value=0)

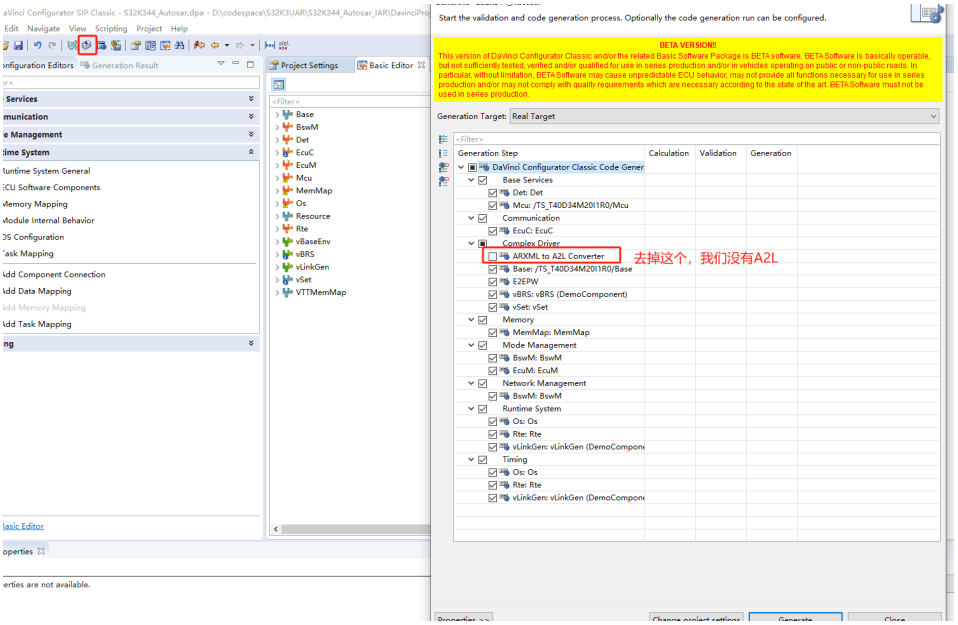
RAM

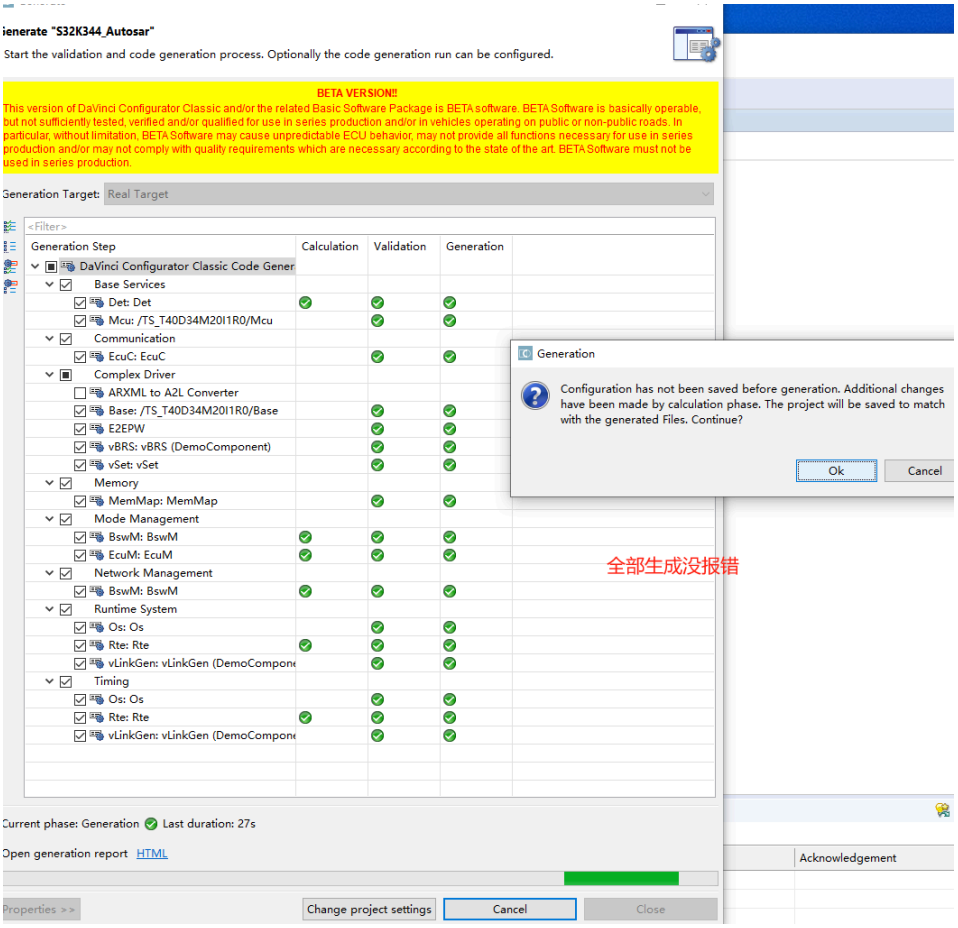




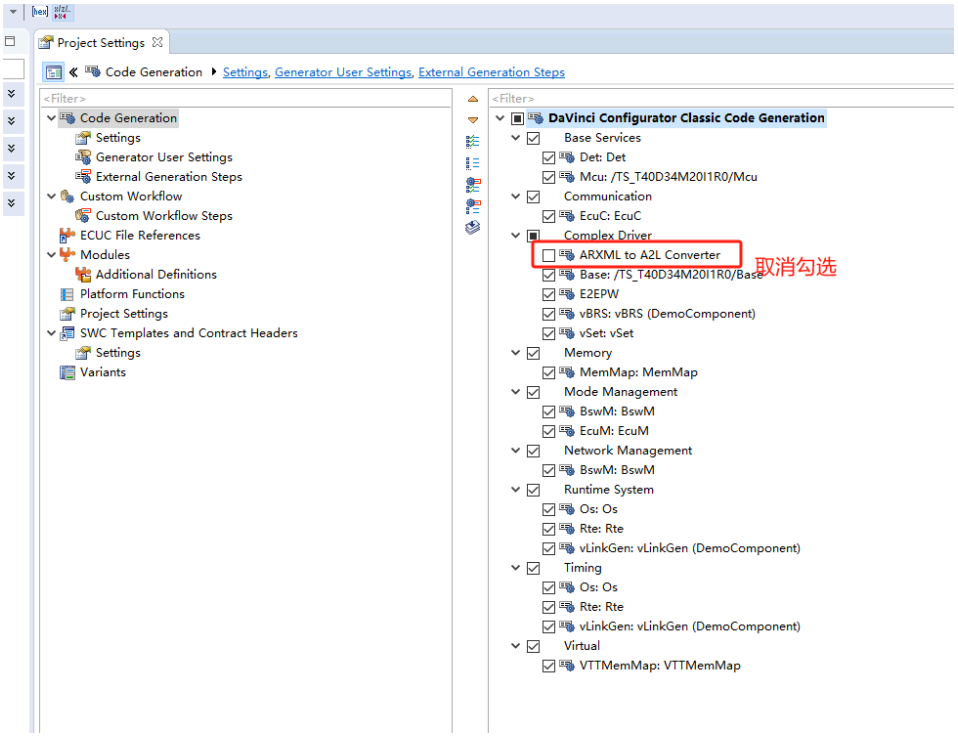
三、生成模块

注意MCAL模块Base和Mcu需要依赖EB的license, 参考文章“EB+MCAL(RTD)+S32DS+JLink开发环境安装（基于S32K344）”的2.1节





取消勾选，这样生成模块就默认不勾选ARXML to A2L Converter


Project Settings
Code Generation
Settings, Generator User Settings, External Generation Steps
<Filter>
Code Generation
Settings
Generator User Settings
External Generation Steps
Custom Workflow
Custom Workflow Steps
ECUC File References
Modules
Additional Definitions
Platform Functions
Project Settings
SWC Templates and Contract Headers
Settings
Variants
<Filter>
DaVinci Configurator Classic Code Generation
Base Services
Det: Det
Mcu: /TS_T40D34M20I1R0/Mcu
Communication
EcuC: EcuC
Complex Driver
ARXML to A2L Converter
取消勾选
Base: /TS_T40D34M20I1R0/Base
E2EPW
vBRS: vBRS (DemoComponent)
vSet: vSet
Memory
MemMap: MemMap
Mode Management
BswM: BswM
EcuM: EcuM
Network Management
BswM: BswM
Runtime System
Os: Os
Rte: Rte
vLinkGen: vLinkGen (DemoComponent)
Timing
Os: Os
Rte: Rte
vLinkGen: vLinkGen (DemoComponent)
Virtual
VTTMemMap: VTTMemMap